# The Complete Treatise on Cyclic Quotient Systems

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**Abstract**

This treatise develops the algebra, statistical foundations, and substantive interpretations of cyclic quotient systems. A cyclic quotient system is a finite collection of nonzero quantities arranged in a directed cycle, with each edge representing a quotient of adjacent quantities. The defining identity is a telescoping closure condition: the product of the quotients around the cycle is one. The algebra is then generalized to log coordinates, incidence matrices, graphs, and graphs with noisy or partially observed data. The resulting formalism is then assessed mathematically, statistically, econometrically, and in economic, financial, scientific, and civic contexts. The central principle is strict: algebraic closure is a model of consistency, not an empirical law or a causal mechanism. The treatise emphasizes that a quotient system can structure evidence, reveal inconsistency, or encode relative information, but it does not by itself establish causation, welfare superiority, legal validity, or physical law. All major conclusions are derived in an explicit and transparent way.

The treatise ends with "The End"

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## 1 Introduction

Let $x_1, \dots, x_N$ be nonzero quantities. A cyclic quotient system assigns to each adjacent pair a quotient of the form

$$q_i = \frac{x_i}{x_{i+1}}, \qquad i = 1, \dots, N, \tag{1}$$

with the cyclic convention $x_{N+1} = x_1$. The basic identity is

$$\prod_{i=1}^{N} q_i = \frac{x_1}{x_2}\frac{x_2}{x_3} \cdots \frac{x_N}{x_1} = 1. \tag{2}$$

This is a telescoping identity. It is exact whenever every denominator is nonzero.

The first objective of this treatise is to derive the algebraic structure of such systems in its generality. The second is to build a statistical framework for noisy and repeated observations. The third is to connect the formalism to economics, welfare analysis, financial planning, econometrics, law, and public policy without blurring the boundary between algebraic identity and empirical inference. Finally, the treatise explains the relation between quotient closure and graph theory, linear algebra, and interdisciplinary modeling.

The thesis is deliberately modest but rigorous: cycle closure is a consequence of construction, not a substantive conclusion. It is a structural invariant. It can be used to design, validate, and interpret measurements, but it cannot by itself establish a model of behavior, a market mechanism, a causal effect, or a law of nature. The mathematical and conceptual burden of interpretation remains with the researcher or analyst.

# 2 Foundations of cyclic quotient algebra

## 2.1 Definition and basic theorem

**Definition 1.** *A cyclic quotient system of order $N$ is a tuple*

$$\mathcal{C} = (x_1, \dots, x_N), \tag{3}$$

*where each $x_i \neq 0$, together with the directed edge quotients*

$$q_i = \frac{x_i}{x_{i+1}}, \qquad i = 1, \dots, N, \qquad x_{N+1} = x_1. \tag{4}$$

**Theorem 1** (Cyclic closure)**.** *For every cyclic quotient system,*

$$\prod_{i=1}^{N} q_i = 1. \tag{5}$$

*Proof.* Substituting the quotient definition gives

$$\prod_{i=1}^{N} q_i = \frac{x_1}{x_2}\frac{x_2}{x_3} \cdots \frac{x_{N-1}}{x_N}\frac{x_N}{x_1} \tag{6}$$

$$= 1, \tag{7}$$

since each numerator factor cancels with the matching denominator factor. The proof requires only that each quotient be defined, i.e. each $x_i \neq 0$. □ □

This identity is the fundamental algebraic object. It is exact, finite, and independent of context. It is not an empirical law.

## 2.2 Equivalent forms

The closure identity can be written in many mathematically equivalent ways. A particularly useful one is

$$\sum_{i=1}^{N} \log q_i = 0, \tag{8}$$

provided $x_i > 0$ for all $i$. This becomes

$$\sum_{i=1}^{N} (\log x_i - \log x_{i+1}) = 0. \tag{9}$$

Equivalently,

$$\sum_{i=1}^{N} \log x_i - \sum_{i=1}^{N} \log x_{i+1} = 0. \tag{10}$$

Since the index set is cyclic, the sum of log differences vanishes by cancellation.

### 2.3 The three-variable case

For three positive variables $P_t$, $A_t$, and $G_t$,

$$\frac{P_t}{A_t}\frac{A_t}{G_t}\frac{G_t}{P_t} = 1. \tag{11}$$

This is the canonical example in the document. The identity is correct if all terms are defined and nonzero.

However, it is essential to distinguish:

- an algebraic identity, - a measurement identity, - a substantive economic interpretation, - a causal claim.

The cycle identity is the first. It is not the second or the fourth.

### 2.4 Permutation invariance and orientation

The order of variables is a modeling decision. Let $\pi$ be a permutation of $\{1, \ldots, N\}$. Define

$$q_i = \frac{x_{\pi(i)}}{x_{\pi(i+1)}}, \tag{12}$$

with $\pi(N+1) = \pi(1)$. Then the same proof yields

$$\prod_{i=1}^{N} q_i = 1. \tag{13}$$

Thus the cycle identity is invariant under relabeling and permutation of the node order, provided the same cycle is preserved.

Reversing orientation produces reciprocals:

$$\frac{x_i}{x_{i+1}} \quad \longleftrightarrow \quad \frac{x_{i+1}}{x_i}. \tag{14}$$

This reversal corresponds to traversing the same geometric or conceptual edge in the opposite direction.

### 2.5 Graph-theoretic generalization

A quotient cycle is a special case of an oriented graph. Let $G = (V, E)$ be a finite directed graph with vertices $v \in V$ and directed edges $e \in E$. For each vertex $v$, assign a nonzero quantity $x_v$. For each directed edge $u \to v$, define

$$q_{uv} = \frac{x_u}{x_v}. \tag{15}$$

Then for any closed oriented walk

$$v_0 \to v_1 \to \cdots \to v_m = v_0, \tag{16}$$

the corresponding product of edge quotients satisfies

$$\prod_{k=0}^{m-1} \frac{x_{v_k}}{x_{v_{k+1}}} = 1. \tag{17}$$

This is the general closed-walk identity.

A cycle is therefore a closed walk of minimal interest; a directed graph with multiple cycles induces multiple algebraic closure constraints. The redundancy count is governed by the cycle space dimension.

# 3 Logarithmic representation and linear algebra

## 3.1 The incidence matrix

Let $x_i > 0$ for all $i$, and define

$$z_i = \log x_i. \tag{18}$$

Then the quotient is represented by the log-difference

$$r_i = \log q_i = z_i - z_{i+1}. \tag{19}$$

Summing around the cycle gives

$$\sum_{i=1}^{N} r_i = \sum_{i=1}^{N} (z_i - z_{i+1}) = 0. \tag{20}$$

Let $B$ be the oriented incidence matrix of the cycle graph. Then

$$r = Bz. \tag{21}$$

For the $N$-cycle, $B$ has one row per edge and one column per node. Each row has a $+1$ in the source position and a $-1$ in the target position. Then

$$\mathbf{1}^{\mathsf{T}} B = 0, \qquad B\mathbf{1} = 0. \tag{22}$$

The first identity expresses that the sum of log-quotients around the cycle is zero; the second states that a common additive shift in all log-levels leaves all ratios unchanged. In multiplicative terms, scaling all $x_i$ by the same positive constant does not change ratios.

## 3.2 Identifying the degrees of freedom

The quotient system identifies relative levels only. Given $q_1, \ldots, q_{N-1}$, the vector $x_1, \ldots, x_N$ is determined up to one positive scalar factor. Symbolically, if $x_1$ is chosen, then

$$x_2 = \frac{x_1}{q_1}, \qquad x_3 = \frac{x_2}{q_2}, \qquad \ldots, \qquad x_N = \frac{x_{N-1}}{q_{N-1}}. \tag{23}$$

The final edge then imposes the closure condition

$$\frac{x_N}{x_1} = q_N = \prod_{i=1}^{N-1} q_i^{-1}. \tag{24}$$

Hence the cycle has $N - 1$ independent relative contrasts and one redundancy. The redundant equation is not new information about the scale, only a consistency condition.

## 3.3 Cycle space and redundancy

For a connected graph with $V$ vertices and $E$ edges, the cycle space dimension is

$$\dim(\text{CycleSpace}) = E - V + 1. \tag{25}$$

This number is the number of independent feedback loops. In a quotient system, each independent cycle gives one algebraic closure equation. This does not mean there are $E - V + 1$ independent empirical facts; it only tells us how many independent algebraic constraints are embedded in the graph. The graph may be connected, but the data may still be noisy, incomplete, or inconsistent.

# 4 Changes, growth factors, and rates of change

## 4.1 Arithmetic changes and their cyclic product

Let $x_i(t)$ be observed at time $t$, and define the arithmetic change

$$\Delta x_i = x_i(t+1) - x_i(t). \tag{26}$$

If every $\Delta x_i \neq 0$, then the change quotient system satisfies

$$\prod_{i=1}^{N} \frac{\Delta x_i}{\Delta x_{i+1}} = 1, \qquad \Delta x_{N+1} = \Delta x_1. \tag{27}$$

This follows immediately from the same closure identity applied to the numbers $\Delta x_1, \ldots, \Delta x_N$.

**Example 1.** *For $P_t, A_t, G_t$,*

$$\frac{\Delta P_t}{\Delta A_t}\frac{\Delta A_t}{\Delta G_t}\frac{\Delta G_t}{\Delta P_t} = 1. \tag{28}$$

*This perfectly valid identity is not an elasticity, not a marginal effect, and not a causal law. It is a statement about the arithmetic changes themselves.*

## 4.2 Growth factors and log growth

For positive data, define growth factors

$$g_i = \frac{x_i(t+1)}{x_i(t)}. \tag{29}$$

Then

$$\prod_{i=1}^{N} \frac{g_i}{g_{i+1}} = 1 \tag{30}$$

by the same telescoping property. Equivalently, in log terms,

$$\sum_{i=1}^{N} (\log g_i - \log g_{i+1}) = 0. \tag{31}$$

This is different from arithmetic-change closure because

$$\frac{x_i(t+1) - x_i(t)}{x_{i+1}(t+1) - x_{i+1}(t)} \neq \frac{x_i(t+1)/x_i(t)}{x_{i+1}(t+1)/x_{i+1}(t)} \tag{32}$$

in general. Different forms of change answer different questions.

## 4.3 The danger of zero and near-zero denominators

The quotient identity requires domains where the denominators are defined and nonzero. For arithmetic changes, if $\Delta x_i = 0$, the ratio $\Delta x_i / \Delta x_{i+1}$ may be undefined or numerically unstable. A cycle with one zero change does not permit the same closure calculation without qualification.

Therefore, the analyst must report:

1. the exact denominator values,
2. zero and near-zero diagnostics,
3. the time interval,
4. unit transformations,
5. sensitivity to outlier adjustment.

A ratio can explode near zero even when all variables are finite. This is not a mathematical contradiction; it is a computational and interpretive warning.

# 5 Statistical foundations

## 5.1 Measurement model

Suppose the true levels are $x_i$ and the observed values are $\widetilde{x}_i$. The measured quotients are

$$\widetilde{q}_i = \frac{\widetilde{x}_i}{\widetilde{x}_{i+1}}. \tag{33}$$

Then there may be a residual discrepancy

$$\varepsilon = \prod_{i=1}^{N} \widetilde{q}_i - 1, \tag{34}$$

or, when using logs,

$$\varepsilon_{\log} = \sum_{i=1}^{N} \log \widetilde{q}_i. \tag{35}$$

This residual can be due to:

- measurement error,
- rounding,
- incompatible dates or spatial units,
- missing or replaced observations,
- misclassification,
- different definitions for different variables.

The closure identity still holds for the underlying true values. The observed values may not.

## 5.2 Constrained estimation under closure

If the observed log-quotients are $\widetilde{r}_i$, and these are noisy estimates of true $r_i$ with

$$\widetilde{r} = r + u, \qquad \mathbf{1}^{\mathsf{T}} r = 0, \tag{36}$$

then the analyst may estimate $r$ by minimizing

$$(\widetilde{r} - a)^{\mathsf{T}} \Sigma^{-1} (\widetilde{r} - a) \tag{37}$$

subject to

$$\mathbf{1}^{\mathsf{T}} a = 0. \tag{38}$$

The solution is

$$\widehat{r} = \widetilde{r} - \Sigma \mathbf{1} \frac{\mathbf{1}^{\mathsf{T}} \widetilde{r}}{\mathbf{1}^{\mathsf{T}} \Sigma \mathbf{1}}. \tag{39}$$

This is a projection that enforces exact closure. It is transparent and disciplined. The analyst must disclose the measurement model and the adjustment.

The crucial point is that the closure constraint is not an empirical fact about the world; it is a property of the chosen representation. It is imposed as an exact condition under a model. Without the model, the residual is just evidence of mismatch.

### 5.3 Dependence of adjacent ratios

Adjacent log contrasts share a variable:

$$\log \frac{x_i}{x_{i+1}} \quad \text{and} \quad \log \frac{x_{i+1}}{x_{i+2}}. \tag{40}$$

They are statistically dependent because both contain $\log x_{i+1}$. If one treats ratios as independent observations without accounting for this dependence, standard errors are generally understated. The covariance structure matters.

In a cycle with $N$ nodes, there are only $N - 1$ independent differences in the log-level vector because of the closure restriction. This is the exact algebraic reason why the last edge is redundant. Therefore:

- use a spanning tree for independent contrasts,
- use extra edges for redundancy checks,
- do not count them as independent empirical facts.

### 5.4 Interpretation of residuals

A residual near zero is not evidence of truth; it is evidence of consistency within the chosen representation. A large residual may indicate:

- incompatible measurement units,
- inconsistent dates,
- repeated or rescaled variables,
- denominator instability,
- missing data imputation failure.

The analyst should not infer causality from the residual alone.

## 6 Economics, welfare, and public policy

### 6.1 The population-area-Gini example

Let $P_t$, $A_t$, and $G_t$ denote population, area, and a Gini coefficient, respectively. For positive values, the cyclic quotient system is

$$\frac{P_t}{A_t}\frac{A_t}{G_t}\frac{G_t}{P_t} = 1. \tag{41}$$

The component terms have the following role:

$$\frac{P_t}{A_t} = \text{population density in persons per unit area}, \tag{42}$$

$$\frac{A_t}{G_t} = \text{area relative to inequality summary}, \tag{43}$$

$$\frac{G_t}{P_t} = \text{inequality relative to population size}. \tag{44}$$

The first expression has a standard economic interpretation. The second and third do not by themselves have a canonical economic meaning; they are algebraic contrasts. Their meaning must be justified by a substantive model.

### 6.2 Normalizing and scaling

Suppose $G_t$ is measured on a 0-to-1 scale. The quotient $A_t/G_t$ is undefined if $G_t = 0$. If $G_t$ is measured in percentage points, the ratio changes by a multiplicative factor of 100. Thus, the value of the quotient depends on the chosen scale. The algebra does not determine the correct scale; the measurement protocol does.

Therefore, the analyst must clearly state:

- whether the Gini is on a 0–1 scale or 0–100 scale,
- whether population counts include residents, commuters, migratory workers, or legal residents,
- whether the area variable is land area, urban area, or administrative area,
- whether the time period is annual, monthly, or quarterly.

### 6.3 Arithmetic change interpretation

For changes,

$$\frac{\Delta P_t}{\Delta A_t}\frac{\Delta A_t}{\Delta G_t}\frac{\Delta G_t}{\Delta P_t} = 1. \tag{45}$$

This identity is mathematically exact, but its economic content is limited. A ratio of changes is not necessarily:

- a marginal effect,
- an elasticity,
- a productivity parameter,
- a causal response,
- a welfare measure.

It is only an algebraic consequence of the change variables.

### 6.4 Econometric identification

A legitimate econometric specification may be

$$y_{it} = \alpha_i + \lambda_t + \beta d_{it} + \gamma^{\mathsf{T}} w_{it} + u_{it}, \tag{46}$$

where $y_{it}$ is an outcome such as income, development, urban service access, or welfare, $d_{it}$ is a density or land-use measure, and $w_{it}$ are controls. The cycle identity is not a substitute for the valid economic model. It can be used to check internal consistency of the underlying variables, but it cannot identify $\beta$ without assumptions such as exogeneity, fixed effects, or an instrument.

### 6.5 Welfare economics

An economy may have a Gini coefficient that falls while absolute poverty rises, or vice versa. An increase in population density may correspond to more economic opportunity and also more congestion. A quotient cycle does not define a social welfare ordering. To compare welfare, one must specify:

- a social welfare function,
- distributional weights,

- poverty thresholds,
- ethical assumptions,
- uncertainty and inequality in outcomes.

This is a normative question, not a mathematical consequence of closure.

# 7 Financial planning and analysis

Financial ratios are two-sided by construction: they can reveal patterns but cannot by themselves validate a business model.

A standard balance-sheet example is

$$\frac{\text{Revenue}}{\text{Assets}} \cdot \frac{\text{Assets}}{\text{Debt}} \cdot \frac{\text{Debt}}{\text{Revenue}} = 1. \tag{47}$$

This identity is mathematically correct whenever all terms are strictly positive. But it is not evidence of solvency, profitability, or good governance. The ratio system is helpful for:

- internal bookkeeping,
- consistency tests,
- scenario analysis,
- dashboard design,
- stress testing.

It is not a substitute for:

- GAAP or IFRS reporting,
- cash-flow analysis,
- forecasting assumptions,
- risk modeling,
- legal compliance.

Financial planning must distinguish accounting identity from economic value, cash flow, and future uncertainty.

# 8 Scientific and interdisciplinary connections

## 8.1 Physics

In physics, a quantity may be expressed as a potential difference, for example,

$$\Delta\phi = \phi_u - \phi_v. \tag{48}$$

In a closed path,

$$\sum_{k=1}^{N} (\phi_{v_k} - \phi_{v_{k+1}}) = 0. \tag{49}$$

This is structurally similar to the log-quotient identity. However, the algebraic resemblance does not imply a physical conservation law. The physical law must be derived from the governing equations or measurement theory. A cycle in positive quantities is not by itself a law of motion or a field equation.

### 8.2 Chemistry

In chemistry, ratios of concentrations, activities, or partial pressures may be analyzed under appropriate equilibrium assumptions. However, the quotient closure identity does not imply chemical equilibrium or reaction kinetics. Those require thermodynamic relations, stoichiometric balances, and often measurable conditions such as temperature and pressure.

### 8.3 Biology

Ecological or biological quantities may be represented as populations or frequencies. Ratios and log ratios can be useful for growth analysis or relative abundance. Yet a cycle algebra does not imply:

- competition,
- mutualism,
- fitness advantage,
- causation,
- selective pressure.

Biological mechanisms require domain-specific theory and evidence.

### 8.4 Linguistics

In linguistics, one may compare token frequencies, lexical density, or dialectal prevalence. Ratios can be useful descriptors. But a ratio system is not a grammar, not a theory of language acquisition, and not a causal explanation of change. Meaningful inference requires corpus design, metadata availability, and theoretical justification.

### 8.5 Psychology and psychiatry

Psychological tests and psychiatric assessments require construct validation. A quotient may be a descriptor of a score pattern, but it is not, by itself, a latent trait or a diagnosis. The ethical and professional requirements of mental health work exceed any algebraic reduction. The responsible protocol is:

- validity,
- reliability,
- privacy,
- informed consent,
- professional review.

### 8.6 Philosophy and religion

The distinction between an identity and an interpretation is central to philosophy of science. A cyclic quotient identity is formal and exact; its application to institutions, meaning, or value systems is not. Religion and metaphysics make claims about ultimate reality and meaning that go beyond formal algebra. The formalism may help structure arguments, but it does not decide metaphysical questions.

### 8.7 Political science and international relations

Political scientists often use ratios such as representation levels, fiscal capacity, or participation rates. These are descriptive summaries. They cannot establish:

- institutional quality,
- democratic legitimacy,
- diplomatic success,
- strategic intent,
- international legal obligations.

International relations requires historical, legal, and strategic analysis. Quantitative indices are aids, not determinants.

### 8.8 Conflict, welfare, and international law

Because war and conflict are high stakes, any quantitative analysis must be framed carefully. A ratio of casualties, mobilization, resource intensity, or displacement can describe patterns, but it cannot justify force, determine morality, or replace legal standards. In humanitarian law, principles such as distinction, proportionality, and precaution are legal and ethical constraints, not calculable outputs of a quotient algebra. This treatise does not serve as operational guidance for violence or coercion.

## 9 Computer science, algorithms, and data science

### 9.1 Cycle checks in code

A generic implementation of a cycle check is:

```
procedure CycleCheck(x[1..N]):
require N ≥ 2
for i = 1 to N:
if x[i] = 0 then return UNDEFINED
product = 1
for i = 1 to N:
j = ((i mod N) + 1)
q[i] = x[i] / x[j]
product = product * q[i]
return product
```

This is $O(N)$ in time and $O(N)$ in memory if all quotients are stored. The same logic can be implemented in log space for positive data:

$$\sum_{i=1}^{N} \log \frac{x_i}{x_{i+1}} = 0. \tag{50}$$

This is usually more stable numerically.

### 9.2 Graph algorithms and weighted least squares

Given log-quotient observations $r$ and an incidence matrix $B$, estimate the underlying log-level vector $z$ by solving

$$\widehat{z} = \arg\min_{z} (r - Bz)^{\mathsf{T}} W (r - Bz), \tag{51}$$

where $W$ is a positive-definite weight matrix. If a constant shift is unidentifiable, impose a normalization such as

$$\mathbf{1}^{\mathsf{T}} z = 0. \tag{52}$$

This is standard in graph-based estimation. It is mathematically coherent but still subject to measurement-model assumptions.

### 9.3 AI and machine learning

A cyclic quotient may be a useful engineered feature or consistency check. For example, a system may compare a ratio of observed inputs to an expected ratio. Yet a feature derived by algebraic closure is not a guarantee of semantic validity. In ML, the essential questions are:

- Is the target meaningful?
- Is the data representative?
- Are there confounders, leakage, or missingness?
- Is the model calibrated and robust?
- Are the documented assumptions valid?

A quotient system can be part of a data pipeline, but it does not replace model validation.

### 9.4 Cybersecurity and defensive computing

In cybersecurity, quotient cycles may serve as a defensive check for consistency among system metrics or replicated observations. But they cannot detect all malice patterns or guarantee system security. A secure system requires:

- identity and access controls,
- authentication and authorization,
- patching and configuration management,
- logging and audit trails,
- incident response,
- data protection and governance.

This treatise intentionally does not provide operational attack guidance or exploit methods.

# 10 Law, governance, privacy, and ethics

The formalism of cyclic quotients is neutral, but applied use is not. Any use of personal or sensitive data requires legal and ethical review. Important questions include:

- lawful basis for data use,
- privacy risk,
- data minimization,
- retention schedule,
- public transparency,
- auditability of methodology,
- explanation for decisions.

A quotient is not a legal standard. It is a tool for transparent quantification, and transparency requires disclosure of definitions, assumptions, and uncertainty.

# 11 Mathematical proofs and derivations

## 11.1 Lemma: product closure under permutation

Given

$$q_i = \frac{x_{\pi(i)}}{x_{\pi(i+1)}}, \tag{53}$$

then

$$\prod_{i=1}^{N} q_i = 1. \tag{54}$$

*Proof.* The product is

$$\prod_{i=1}^{N} \frac{x_{\pi(i)}}{x_{\pi(i+1)}} = \frac{x_{\pi(1)}}{x_{\pi(2)}} \frac{x_{\pi(2)}}{x_{\pi(3)}} \cdots \frac{x_{\pi(N)}}{x_{\pi(1)}} \tag{55}$$

$$= 1. \tag{56}$$

The factors cancel pairwise. □ □

## 11.2 Lemma: loop sum of log differences is zero

If $x_i > 0$, then for the cycle,

$$\sum_{i=1}^{N} \log \frac{x_i}{x_{i+1}} = 0. \tag{57}$$

*Proof.*

$$\sum_{i=1}^{N} \log \frac{x_i}{x_{i+1}} = \sum_{i=1}^{N} (\log x_i - \log x_{i+1}) \tag{58}$$

$$= \sum_{i=1}^{N} \log x_i - \sum_{i=1}^{N} \log x_{i+1} \tag{59}$$

$$= 0, \tag{60}$$

because the two sums are over the same indices in the same cyclic order. □ □

### 11.3 Proposition: common scalar invariance

If $c > 0$, then replacing each $x_i$ by $cx_i$ leaves every quotient unchanged.

*Proof.*

$$\frac{cx_i}{cx_{i+1}} = \frac{x_i}{x_{i+1}}. \tag{61}$$

The factor $c$ cancels. □ □

This is the formal reason why the quotient system identifies relative values, not an absolute scale.

### 11.4 Corollary: graph quotient closure on closed walks

For any closed walk $v_0, \ldots, v_m = v_0$,

$$\prod_{k=0}^{m-1} \frac{x_{v_k}}{x_{v_{k+1}}} = 1. \tag{62}$$

*Proof.* This is the same telescoping cancellation as before. □ □

## 12 A compact treatise table of core identities

| Object | Identity | Conditions |
|---|---|---|
| Cycle product | $\prod_{i=1}^{N} \frac{x_i}{x_{i+1}} = 1$ | $x_i \neq 0$ |
| Log cycle sum | $\sum_{i=1}^{N} \log \frac{x_i}{x_{i+1}} = 0$ | $x_i > 0$ |
| Change cycle | $\prod_{i=1}^{N} \frac{\Delta x_i}{\Delta x_{i+1}} = 1$ | $\Delta x_i \neq 0$ |
| Growth cycle | $\prod_{i=1}^{N} \frac{g_i}{g_{i+1}} = 1$ | $g_i > 0$ |
| Graph closed walk | $\prod_{e \in W} q_e = 1$ | each edge quotient defined |

Table 1: Core identities in cyclic quotient systems.

## 13 Summary and conclusion

This treatise has shown that the central object of a cyclic quotient system is a telescoping identity. The product of directed ratios around a closed cycle is one. In log form, the sum of the edge log-differences is zero. In graph language, the quotient system is a linear representation of edge contrasts on a cycle or graph. This is a structural fact with broad mathematical relevance.

It is crucial to keep the algebra separated from interpretation:

- The algebra tells us that the cycle closes.
- The measurement model tells us whether the observed data are consistent.
- The statistical model tells us about uncertainty and dependence.
- The economic or scientific model tells us whether the ratios matter for the research question.
- The causal design tells us whether a relationship can be interpreted as causal.

The algebra is necessary for the representation, but not sufficient for any substantive conclusion.

A cyclic quotient system is thus best viewed as a disciplined way to organize relative information and find inconsistency. It is not a universal theory of behavior, welfare, law, or physical reality. Used carefully, it can lower ambiguity and sharpen analysis. Used carelessly, it can confuse a tautology of closure for explanatory power.

The complete treatise therefore ends with the same principle with which it began: algebraic structure is rigorous, but interpretation requires domain-specific assumptions and evidence. At that boundary, mathematics is not a substitute for theory, ethics, law, or science.

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## Glossary

**Arithmetic change**
The difference $\Delta x_i = x_i(t+1) - x_i(t)$.

**Cycle**
A closed directed sequence of nodes and edges.

**Cycle closure**
The identity that the product of edge ratios around a cycle equals one.

**Cycle space**
The linear space of closed flows on a graph.

**Cyclic quotient system**
A set of nonzero quantities arranged in a closed quotient cycle.

**Growth factor**
: The ratio $g_i = x_i(t+1)/x_i(t)$.

**Incidence matrix**
: A matrix encoding the orientation of edges in a graph.

**Log contrast**
: A difference of log values, such as $\log x_i - \log x_{i+1}$.

**Measurement model**
: A statistical description of how observations relate to latent variables.

**Quotient**
: A ratio $x/y$ defined when $y \neq 0$.

**Residual**
: The discrepancy between observed and exact closure.

**Spanning tree**
: A connected acyclic subgraph.

**Welfare criterion**
: A normative rule for comparing social outcomes.

## The End